

Name: Ankit Pardeshi	Roll no: B-12
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Case Study: Cloud Computing Model of Amazon

Launched in 2006, Amazon Web Services (AWS) fundamentally altered the trajectory of enterprise IT. While many assume AWS was created simply to monetize idle servers from Amazon's retail business, it was actually engineered out of necessity. Amazon needed a standardized, highly automated infrastructure to manage its own internal complexities. Once perfected, this architecture was commercialized to the public.

The Cloud Utility Model

At the heart of AWS is a shift from traditional IT ownership to a "pay-as-you-go" utility model. Instead of buying hardware, businesses rent computing power.

This is delivered through three core layers:

- **Infrastructure as a Service (IaaS):** The underlying building blocks, offering virtual servers (Amazon EC2), storage (Amazon S3), and networking over the internet.
- **Platform as a Service (PaaS):** A tier that abstracts server and OS management, letting developers focus solely on writing and deploying their code.
- **Software as a Service (SaaS):** Ready-to-use, cloud-hosted applications managed end-to-end by AWS.

This model relies on a massive global footprint of distinct Regions and Availability Zones to ensure continuous uptime.

The Proving Ground: Amazon.com

Before dominating the public market, AWS had to solve massive bottlenecks for its own parent company. As Amazon.com's retail operations exploded, its internal infrastructure hit a breaking point.

The Tape Backup Bottleneck

Amazon relied on massive Oracle databases to run its e-commerce platform. Historically, backing up these databases meant relying on physical magnetic tape, a process fraught with friction:

1. **Inefficient Capital Spend:** The company spent millions on physical server space, tape robots, and software licenses.
2. **Inflexible Planning:** Engineers had to forecast and purchase physical storage capacity 12 months ahead of time.
3. **Hazardous Downtime:** If a critical database went down, finding, loading, and reading physical tapes took between 10 and 15 hours—an unacceptable risk for an online retailer.

The Shift to Amazon S3

Amazon's engineering teams bypassed physical hardware entirely by routing all database backups directly to **Amazon S3 (Simple Storage Service)**. S3 provided infinite, API-driven object storage in the cloud.

The Business Impact

The internal transition to cloud infrastructure yielded massive operational wins:

- **12x Faster Recovery:** The time to restore a massive 3.8-terabyte database dropped from 15 hours on tape to just 2.5 hours on S3.
- **On-Demand Storage:** The need for 12-month capacity planning was eliminated, replaced by instant, pay-per-use scalability.

- **Talent Reallocation:** Elite database engineers stopped functioning as hardware technicians and returned to optimizing high-level database architecture.

Global Industry Impact

Amazon's internal success proved that the cloud could handle the world's most demanding workloads. This model quickly became the gold standard for global business. Streaming giant Netflix spent seven years migrating its entire infrastructure to AWS, and Capital One became the first major U.S. bank to go "all-in" on the cloud, leveraging the exact same scalability and agility that first solved Amazon.com's internal hurdles.